4 Muon selections and performance

The muon reconstruction algorithm starts either by finding tracks in the muon detectors, which are then fitted together with tracks reconstructed in the silicon tracker (*global muons*) or by extrapolating tracks from the silicon tracker to match a hit on at least one segment of the muon detectors (*tracker muons*). Muons in the kinematic range $(\mu) > 3\,\mathrm{GeV}/c$, $|\eta(\mu)| < 2.4$ are selected, which allows J/ψ mesons to be well measured within the kinematic range of $1.4 < |y_{\mathrm{J}}/\psi| < 2.4$ and $p_{\mathrm{T}} > 0.2\,\mathrm{GeV}/c$.

The muon candidates are further required to be particle flow muon candidates [42] and pass the *Soft Muon* ID selection, as endorsed by the Muon Physics Object Group (POG) for Run 2 analyses, which can be found here: (`https://twiki.cern.ch/twiki/bin/view/CMS/SWGuideMuonIdRun2#Soft_Muon`). The complete list of all the cuts applied on muons are as follows:

- muons are reconstructed as *particle flow muon candidate*;
- soft muon ID criteria:
  - TMOneStationTight (requires at least one well matched muon segment in any muon station for the track detected in the tracker);
  - the number of tracker layers with hits must be > 5, to ensure sufficient transverse momentum resolution of muons;
  - the number of pixel layers with valid hits > 0, to suppress muons from decays in flight;
  - the track high purity flag is used to reject bad quality tracks, which consists of a few outliers;
  - the distance between the primary event vertex and the muon track in the transverse plane, $D_{xy} < 0.3\,\mathrm{cm}$, and the longitudinal plane, $D_z < 20.0\,\mathrm{cm}$.

Combined acceptance and efficiency of single reconstructed muons with the above selections are studied using PYTHIA generator containing a J/ψ meson decaying into $\mu^+\mu^-$. A gen-level muon particle is considered to be reconstructed if it is matched to a reconstructed-level muon (with the above selections) within an angular cone size of $\Delta R < 0.03$. The combined acceptance and efficiency in 2D as a function of muon η and p_{T}, and in 1D as a function of muon p_{T} averaged over $1.4 < |\eta| < 2.4$ are shown in Fig. 12. The combined acceptance and efficiency in 2D as a function of muon η and total momentum is also shown in Fig. 13.

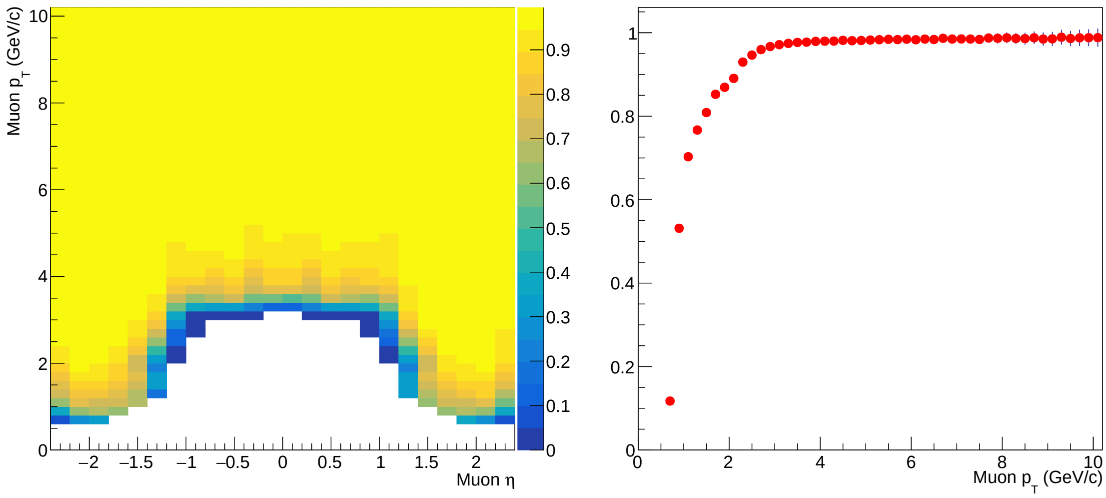

Figure 12: The combined acceptance and efficiency for single muons in 2D as a function of muon η and p_{T} (left), and in 1D as a function of muon p_{T} averaged over $1.4 < |\eta| < 2.4$ (right).

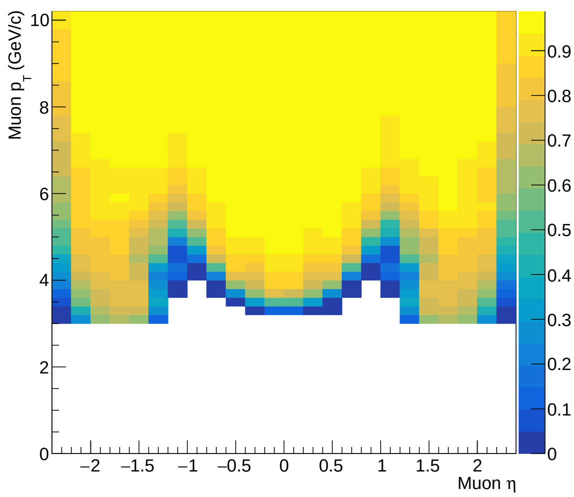

Figure 13: The combined acceptance and efficiency for single muons in 2D as a function of muon η and total momentum.

5 Reconstruction of J/ψ candidates

J/ψ candidates are reconstructed by combining pairs of oppositely charged PF muon candidates, fulfilling the soft muon ID selections as described in Sec. 4, within an invariant mass window within 0.6 GeV/c^2 from the nominal J/ψ mass from particle data group (PDG). The secondary (decay) vertex is reconstructed by using a ROOT "KinematicParticleVertexFitter", where least-mean squared minimization is implemented. In order to suppress combinatorial backgrounds, the vertex fit probability is required to be larger than 1%. The J/ψ kinematic range of $1.4 < |y| < 2.4$ and $0.2 < p_{\mathrm{T}} < 10\,\mathrm{GeV}/c$ is used in the correlation analysis later, to ensure good acceptance and efficiency

Combined acceptance and efficiency of reconstructed J/ψ mesons are studied using PYTHIA generator containing a J/ψ meson decaying into $\mu^+\mu^-$, by matching daughter muons at generator- and reconstructed-level, as shown in Fig. 14. An efficiency of 30–40% can be achieved at low p_{T} ($< 3\,\mathrm{GeV}$) for forward region ($1.4 < |y| < 2.4$).

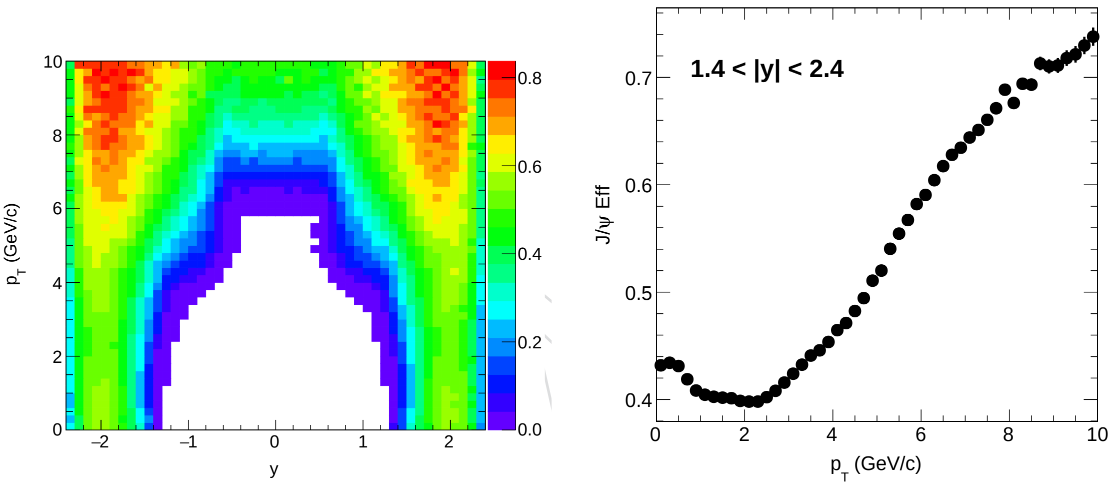

Figure 14: The combined acceptance and efficiency for J/ψ meson in 2D as a function of y and p_{T} (left), and in 1D as a function of p_{T} averaged over $1.4 < |y| < 2.4$ (right).

5.1 Signal extraction

In this analysis, a mass spectrum is always needed to get J/ψ v_2, as described in Section 6.2. The fit procedure and model is studied in details in HIN-16-004 (AN-16-067). The fit function consists of the following components:

- Weighted sum of two Crystal Ball functions $g_{\mathrm{CB}}(m_{\mu^+\mu^-})$ ($f \cdot g_{\mathrm{CB}_1}(m_{\mu^+\mu^-}) + (1-f) \cdot g_{\mathrm{CB}_2}(m_{\mu^+\mu^-})$) to model the J/ψ signal. The two Crystal Ball functions have common mean m_0 and tail parameters α and n.
- 3rd order polynomial to model the combinatorial background.

The Crystal Ball function $g_{\mathrm{CB}}(m)$ combines a Gaussian core and a power-law tail with an exponent n to account for energy loss due to final-state photon radiation, and a parameter α which defines the transition between the Gaussian and the power-law functions,

$$g_{\mathrm{CB}}(m) = \begin{cases} \frac{N}{\sqrt{2\pi}\,\sigma_{\mathrm{CB}}} \exp\left(-\frac{(m-m_0)^2}{2\,\sigma_{\mathrm{CB}}^2}\right), & \text{for } \frac{m-m_0}{\sigma_{\mathrm{CB}}} > -\alpha; \\ \frac{N}{\sqrt{2\pi}\,\sigma_{\mathrm{CB}}} \left(\frac{n}{|\alpha|}\right)^n \exp\left(-\frac{|\alpha|^2}{2}\right) \left(\frac{n}{|\alpha|} - |\alpha| - \frac{m-m_0}{\sigma_{\mathrm{CB}}}\right)^{-n}, & \text{for } \frac{m-m_0}{\sigma_{\mathrm{CB}}} \le -\alpha. \end{cases} \quad (1)$$

The Crystal Ball tail parameters obtained from the J/ψ MC simulations are shown in Tabs. 5. The tail parameters α and n are fixed to the values found in MC (p_{T}-centrality integrated fits) in data fits.

Table 5: Signal shape parameters obtained in J/ψ MC for the double Crystal Ball function.

p_{T} [GeV/c]	α	n
0–1.8	4.31	1.89
1.8–3.0	3.51	1.98
3.0–4.5	3.03	2.03
4.5–6.0	2.74	2.07
6.0–8.0	2.38	2.11
8.0–10.0	2.11	2.15

Figure 15 shows mass spectrum fit for pPb collisions with $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$.

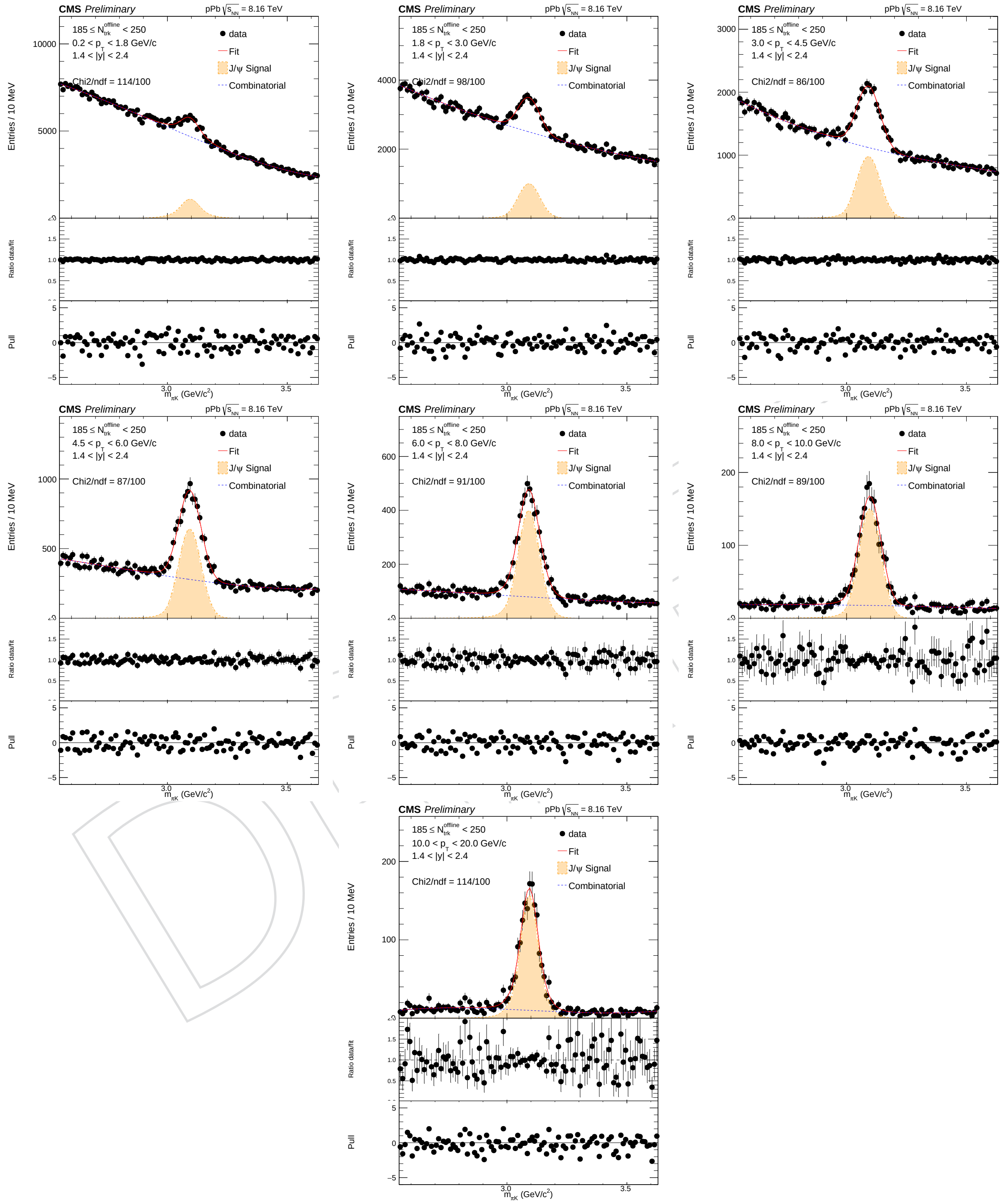

Figure 15: Mass spectrum fit in different p_T bins for pPb collisions with $185 \leq N_{trk}^{offline} < 250$.

5.2 Non-Prompt J/ψ rejection

The procedure of rejecting non-prompt J/ψ has been studied in detail in HIN-16-004 (AN-16-067). Selected information are provided in this section.

5.2.1 The pseudo-proper decay length $\ell^{3D}_{J/\psi}$

The J/ψ mesons coming from b-hadron decays (non-prompt charmonia) are identified by the measurement of a secondary $\mu^+\mu^-$ vertex displaced from the primary collision vertex. The displacement vector between the $\mu^+\mu^-$ vertex and the primary vertex is measured taking into account all directions: x, y and z. The non-prompt charmonia are separated by cutting on pseudo-proper decay length,

$$\ell^{3D}_{\mathrm{J}/\psi} = L_{xyz} \cdot m/p, \tag{2}$$

where m is the PDG J/ψ mass, assumed for all dimuons. This quantity $\ell^{3D}_{\mathrm{J}/\psi}$ is computed from L_{xyz}, where L_{xyz} is the most probable b-hadron decay length in the laboratory frame [43, 44].

In this analysis, we measure prompt J/ψ v_2 by rejecting dimuons with $\ell^{3D}_{\mathrm{J}/\psi}$ above a certain value.

5.2.2 Definition of the cut on the pseudo-proper b-hadron decay length

The $\ell^{3D}_{\mathrm{J}/\psi}$ cut is tuned based on the efficiency of keeping (rejecting) prompt (non-prompt) J/ψ as performed in HIN-16-004 (AN-16-067), where the efficiency is meant for the fraction of events passing a certain $\ell^{3D}_{\mathrm{J}/\psi}$ cut. The strategy is to remove as many non-prompt J/ψ as possible, while keeping the efficiency for prompt J/ψ high. The cut is chosen for each bin in the analysis in such a way that the efficiency for prompt J/ψ is fixed at 90%, which corresponds to a rejection of non-prompt J/ψ of 70–90% or more, as shown in Sec. 5.2.3.

The distributions of $\ell^{3D}_{\mathrm{J}/\psi}$ and an illustration of how we determine the cut values are shown in Fig. 16 for $3 < p_{\mathrm{T}} < 4.5\,\mathrm{GeV}/c$. Since the cut values vary with the J/ψ p_{T}, we calculated them in different p_{T} bins and fitted the obtained pattern by the function $\ell^{3D}_{\mathrm{J}/\psi}(p_{\mathrm{T}}) = a + b/p_{\mathrm{T}} + c * p_{\mathrm{T}} + d * p_{\mathrm{T}}^2$. Figure. 17 shows the dependence of the cut values on p_{T} and the fitted function used for the parametrisation. It is important to note that uncertainties on this parametrisation are irrelevant, and are therefore not given. Then in the analysis the prompt J/ψ are obtained cutting on $\ell^{3D}_{\mathrm{J}/\psi}$, which has to be smaller than the value estimated from the J/ψ p_{T} from the aforementioned $a + b/p_{\mathrm{T}} + c * p_{\mathrm{T}} + d * p_{\mathrm{T}}^2$ function.

Table. 6 provides the fraction of J/ψ signal (obtained from fits to the mass spectrum) passing the $\ell^{3D}_{\mathrm{J}/\psi}$ cuts in pPb data for $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$ and $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$.

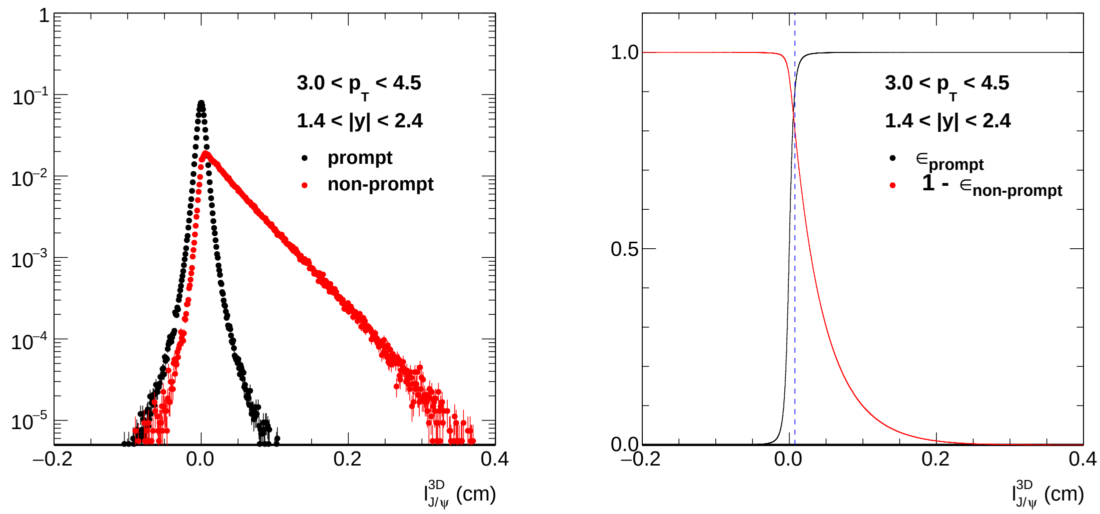

Figure 16: Pseudo-proper decay length distributions of prompt and non-prompt J/ψ in pPb MC simulations, for $3 < p_{\mathrm{T}} < 4.5\,\mathrm{GeV}/c$ and $1.4 < |y| < 2.4$. The right panel shows the prompt efficiency and the non-prompt rejection efficiency, for the chosen $\ell^{3D}_{\mathrm{J}/\psi}$ cut.

Table 6: The fraction of J/ψ signal (obtained from fits to the mass spectrum) passing the $\ell^{3D}_{\mathrm{J}/\psi}$ cuts in pPb data.

p_{T} range (GeV/c)	passing fraction, $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$	passing fraction, $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$
0.2–1.8	85.2%	82.4%
1.8–3.0	79.0%	78.2%
3.0–4.5	75.2%	75.7%
4.5–6.0	76.8%	74.9%
6.0–8.0	74.8%	71.6%
8.0–10.0	71.0%	67.8%

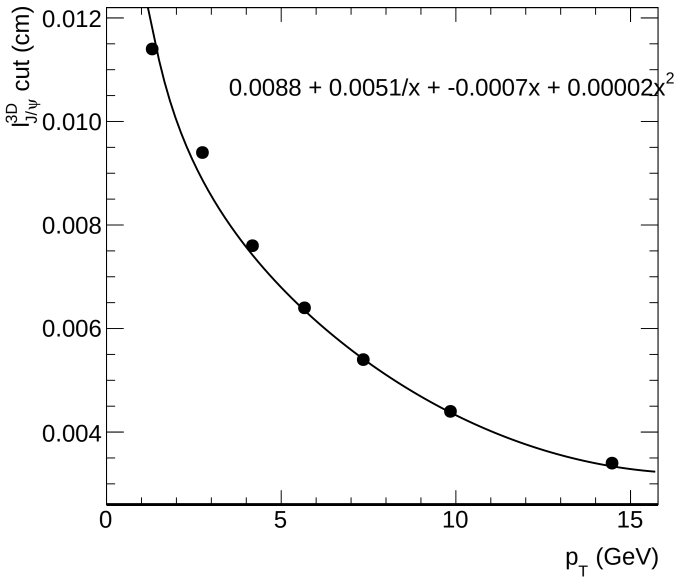

Figure 17: The dependence with p_{T} of the $\ell^{3D}_{\mathrm{J}/\psi}$ cut value giving a 90% efficiency for prompt J/ψ simulation, for $1.4 < |y| < 2.4$. The cut values are fitted by a function ($a + b/p_{\mathrm{T}} + c * p_{\mathrm{T}} + d * p_{\mathrm{T}}^2$). Fit parameters are also shown.

5.2.3 $\ell_{J/\psi}^{3D}$ cut efficiency

The efficiencies of the $\ell_{J/\psi}^{3D}$ cut alone as a function of p_T, can be found in Fig. 18 for MC. By construction, the efficiencies are around 90% for prompt J/ψ. The efficiency for non-prompt J/ψ is found to be up to around 30% at low p_T.

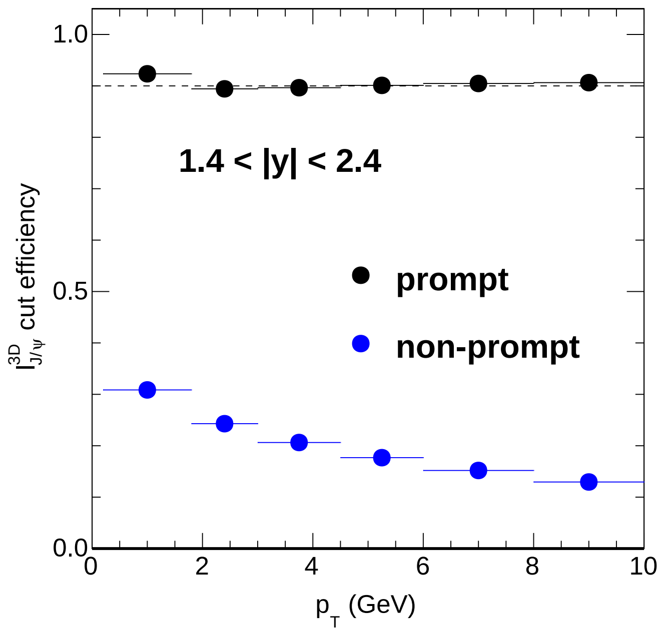

Figure 18: Efficiency of the $\ell_{J/\psi}^{3D}$ cut as a function of p_T in MC, for $1.4 < |y| < 2.4$. A dotted line at 90% has been added to guide the eye.

5.2.4 Comparison of $\ell_{J/\psi}^{3D}$ distribution for J/ψ between data and prompt MC

In order to check the agreement between data and MC for the $\ell_{J/\psi}^{3D}$ distribution and resolution, the $\ell_{J/\psi}^{3D}$ distribution in pPb data and in prompt J/ψ MC are compared in Fig. 19 for $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} 250$ and in Fig. 20 for $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$. For data, candidates from combinatorial background are not expected to follow the MC $\ell_{J/\psi}^{3D}$ distribution. Therefore, better agreement has been observed between data and MC for $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$ than for $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$, and for high p_T than for low p_T.

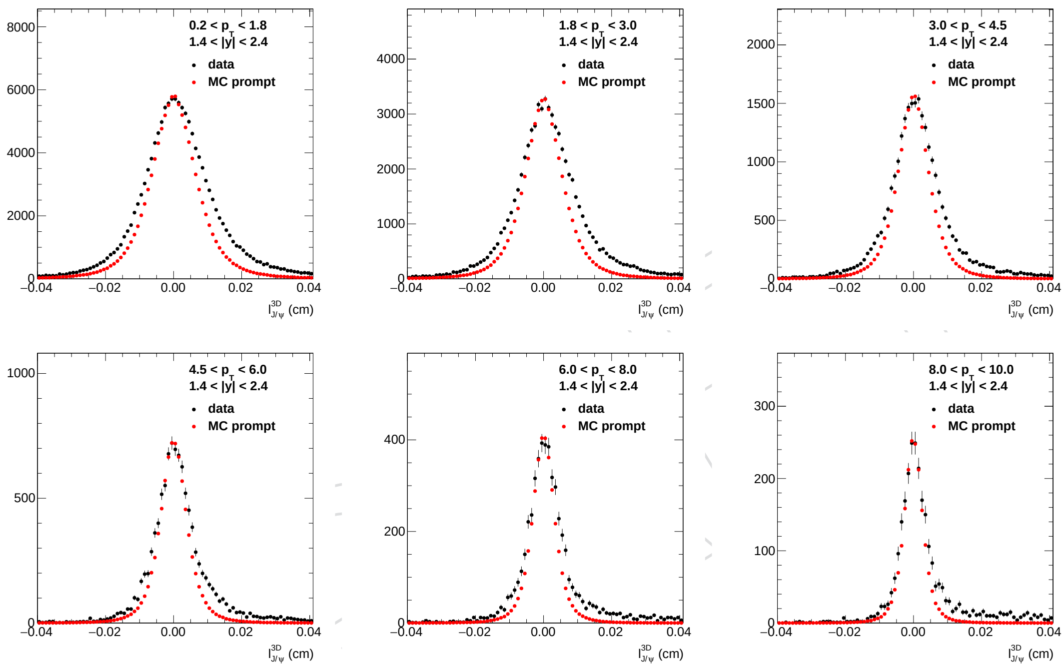

Figure 19: Comparison of the $\ell^{3D}_{J/\psi}$ distribution in pPb data (black) and prompt J/ψ MC (red), for different p_{T} bins for $185 \leq N^{\mathrm{offline}}_{\mathrm{trk}} < 250$. The MC distribution is normalized to the data in the range of $-0.002 < \ell^{3D}_{J/\psi} < 0.002$ cm. All analysis cuts are applied expect the $\ell^{3D}_{J/\psi}$ cut. The J/ψ mass is required to be between 2.9 and 3.25 GeV.

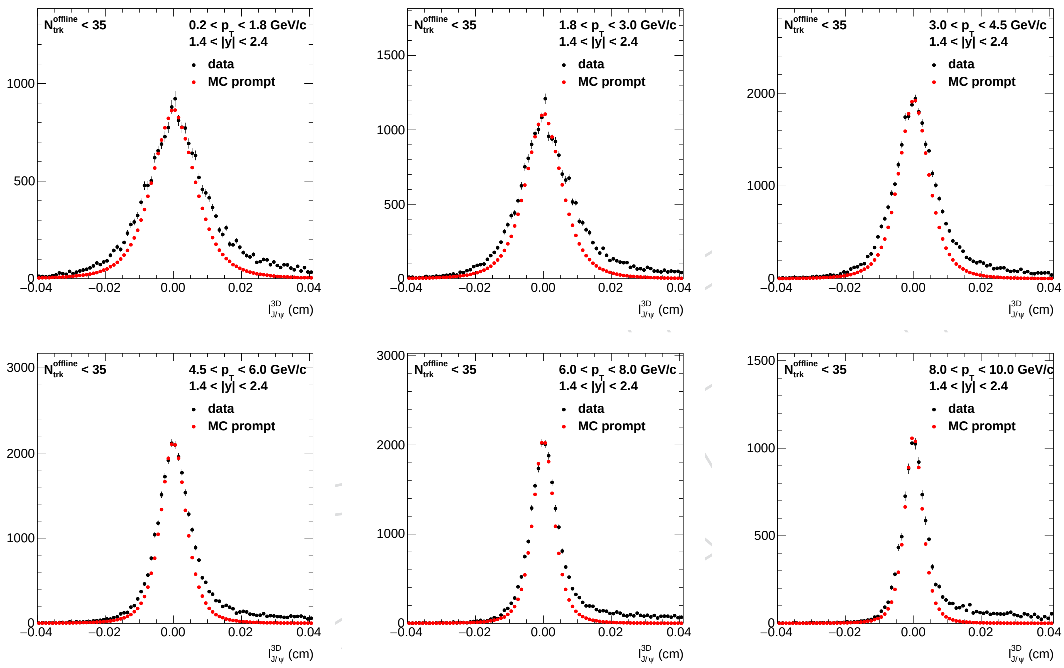

Figure 20: Comparison of the $\ell_{\mathrm{J}/\psi}^{3D}$ distribution in pPb data (black) and prompt J/ψ MC (red), for different p_{T} bins for $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$. The MC distribution is normalized to the data in the range of $-0.002 < \ell_{\mathrm{J}/\psi}^{3D} < 0.002$ cm. All analysis cuts are applied expect the $\ell_{\mathrm{J}/\psi}^{3D}$ cut. The J/ψ mass is required to be between 2.9 and 3.25 GeV.

5.2.5 Unfolding $\ell^{3D}_{J/\psi}$ distribution for signal J/ψ

In order to get rid of effect from the background candidates in the comparison of $\ell^{3D}_{J/\psi}$ between data and MC, data $\ell^{3D}_{J/\psi}$ distributions have been unfolded to get the $\ell^{3D}_{J/\psi}$ for J/ψ signal using RooStats::SPlot. The $\ell^{3D}_{J/\psi}$ distribution in pPb data for signal J/ψ and in prompt J/ψ MC are compared in Fig. 21 for $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} 250$ and in Fig. 22 for $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$. A good agreement has been found for $\ell^{3D}_{J/\psi}$ distributions in data and in MC for prompt J/ψ, while there are contributions from non-prompt J/ψ in data.

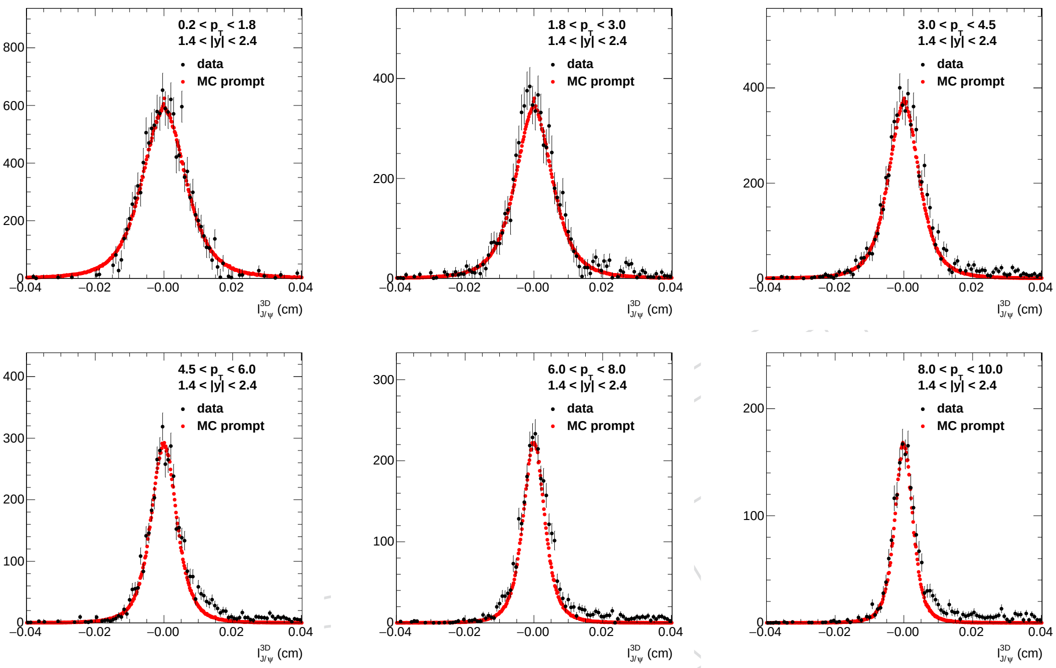

Figure 21: Comparison of the $\ell^{3D}_{J/\psi}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and prompt J/ψ MC (red), for different p_{T} bins for $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$. The MC distribution is normalized to the data in the range of $-0.004 < \ell^{3D}_{J/\psi} < 0.004$ cm. All analysis cuts are applied expect the $\ell^{3D}_{J/\psi}$ cut.

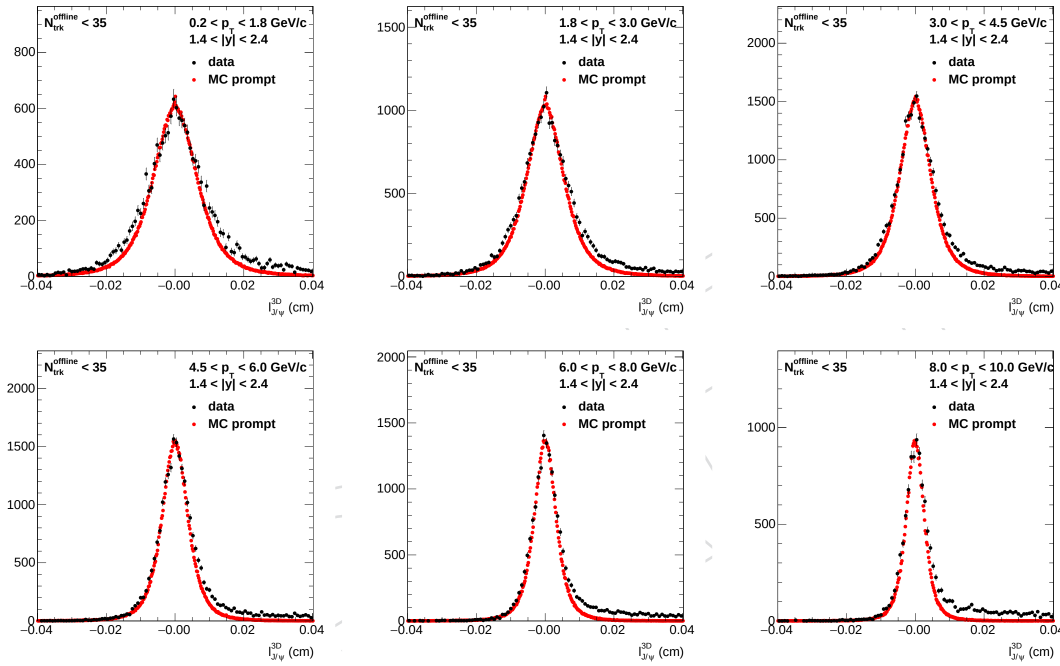

Figure 22: Comparison of the $\ell^{3D}_{J/\psi}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and prompt J/ψ MC (red), for different p_T bins for $N_{\rm trk}^{\rm offline} < 35$. The MC distribution is normalized to the data in the range of $-0.004 < \ell^{3D}_{J/\psi} < 0.004$ cm. All analysis cuts are applied expect the $\ell^{3D}_{J/\psi}$ cut.

5.2.6 Imbalance of $\ell^{3D}_{J/\psi}$ distribution for non-prompt J/ψ in data and MC

In order to check if the non-prompt J/ψ $\ell^{3D}_{J/\psi}$ distributions in data and MC are the same, i.e. the non-prompt rejection efficiency is same in data and MC, a check on the imbalance of $\ell^{3D}_{J/\psi}$ distribution for non-prompt J/ψ has been performed. The negative part ($\ell^{3D}_{J/\psi} < 0$) of the $\ell^{3D}_{J/\psi}$ distributions in Fig. 21 has been reflected with respect to $\ell^{3D}_{J/\psi} = 0$, and then subtracted from the positive part of the $\ell^{3D}_{J/\psi}$ distributions. Since the prompt J/ψ $\ell^{3D}_{J/\psi}$ distributions are symmetric, after the subtraction there is no contribution from prompt J/ψ left. The result distributions represent the imbalance of $\ell^{3D}_{J/\psi}$ distributions for non-prompt J/ψ. The comparison of those distributions can reveal possible difference between non-prompt J/ψ $\ell^{3D}_{J/\psi}$ distributions in data and MC. Fig. 23 and 24 show the comparison for all p_T bins, where within uncertainties no difference has been observed between data and MC. Furthermore, Fig. 25 shows the $\ell^{3D}_{J/\psi}$ cut efficiency from the $\ell^{3D}_{J/\psi}$ imbalance distributions, which are identical within uncertainties. Therefore, the $\ell^{3D}_{J/\psi}$ distributions for non-prompt J/ψ is expected to be same in data and MC.

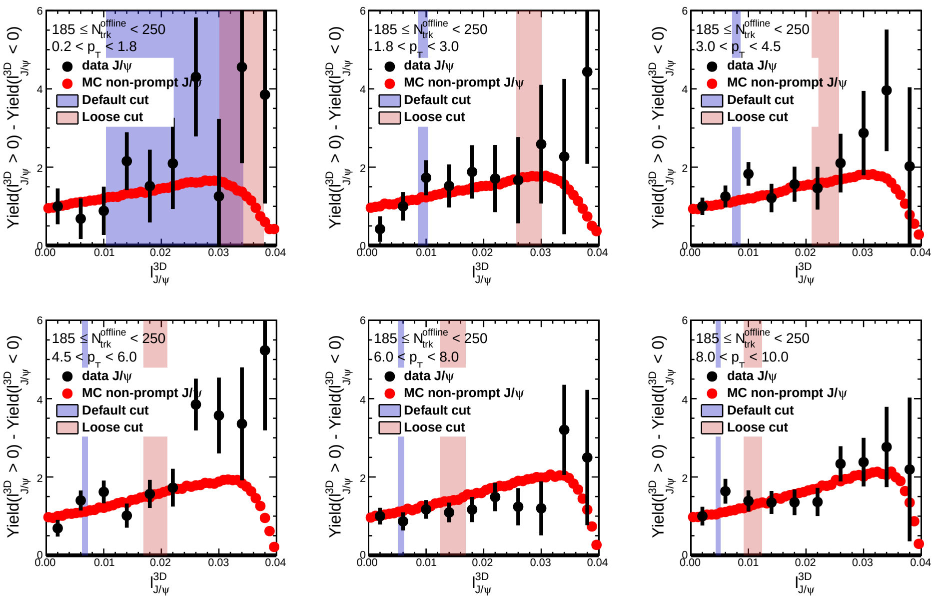

Figure 23: Comparison of the imbalance in $\ell^{3D}_{J/\psi}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and non-prompt J/ψ MC (red), for different p_T bins for $185 \leq N^{\rm offline}_{\rm trk} < 250$. All analysis cuts are applied expect the $\ell^{3D}_{J/\psi}$ cut. The $\ell^{3D}_{J/\psi}$ cuts are indicated by color bands.

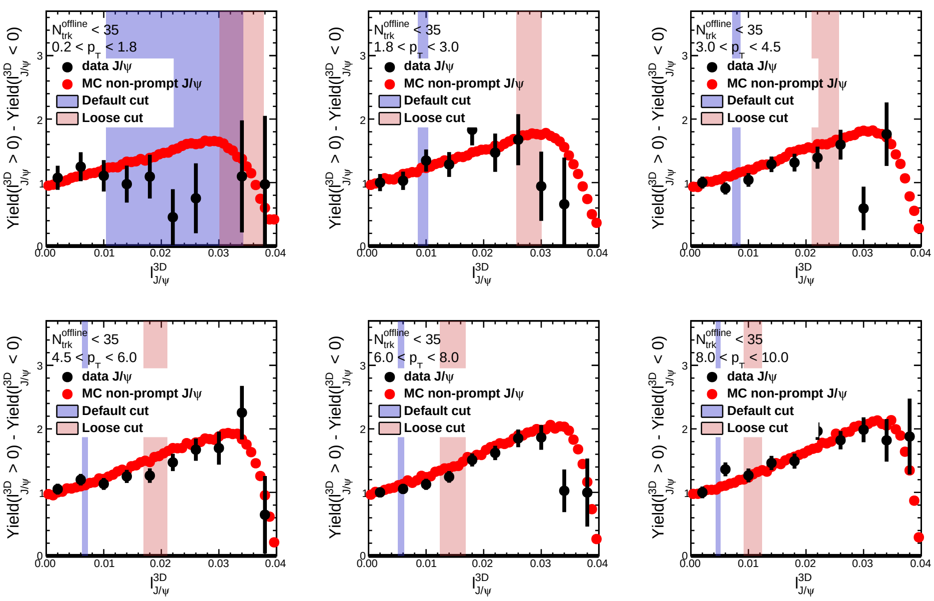

Figure 24: Comparison of the imbalance in $\ell^{3D}_{J/\psi}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and non-prompt J/ψ MC (red), for different p_T bins for $N^{\text{offline}}_{\text{trk}} < 35$. All analysis cuts are applied expect the $\ell^{3D}_{J/\psi}$ cut. The $\ell^{3D}_{J/\psi}$ cuts are indicated by color bands.

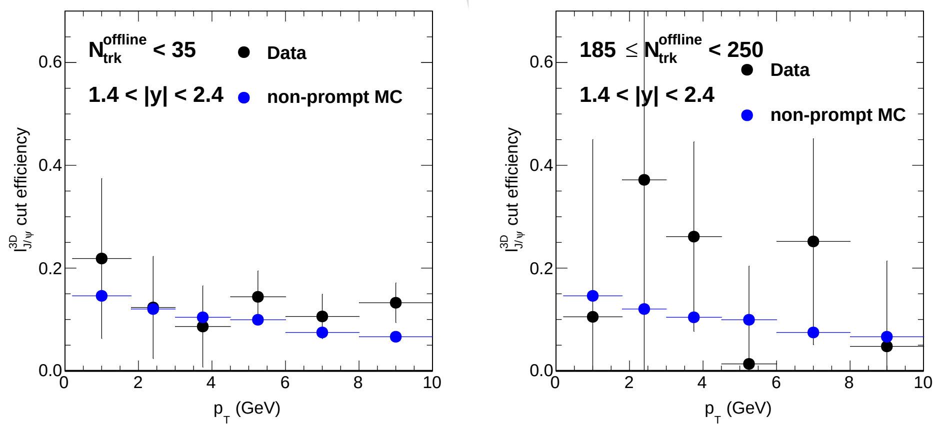

Figure 25: $\ell^{3D}_{J/\psi}$ cut efficiency from the $\ell^{3D}_{J/\psi}$ imbalance distributions in pPb data for signal J/ψ from SPlot unfolding (black) and non-prompt J/ψ MC (blue) for $N^{\text{offline}}_{\text{trk}} < 35$ (left) and $185 \leq N^{\text{offline}}_{\text{trk}} < 250$ (right).

5.2.7 Estimation of non-prompt J/ψ fraction in data before and after $\ell^{3D}_{J/\psi}$ cut

It has been shown in Sec. 5.2.5 and 5.2.6 that $\ell^{3D}_{J/\psi}$ distributions for both prompt and non-prompt J/ψ have good agreement in data and MC, i.e. $\ell^{3D}_{J/\psi}$ cut efficiency in data and MC are identical. Using the $\ell^{3D}_{J/\psi}$ cut efficiency and the fraction of J/ψ passing the $\ell^{3D}_{J/\psi}$ cut in Table. 6, an estimation of non-prompt J/ψ fraction in data is possible.

The fraction of J/ψ passing the $\ell^{3D}_{J/\psi}$ cut (defined as eff_{cut}) can be expressed as combination of prompt and non-prompt J/ψ passing the $\ell^{3D}_{J/\psi}$ cut,

$$eff_{cut} = f_{prompt} * eff_{prompt} + (1 - f_{prompt}) * eff_{non-prompt}, \tag{3}$$

where f_{prompt} is the prompt fraction (before $\ell^{3D}_{J/\psi}$ cut is applied) in data and eff_{prompt} ($eff_{non-prompt}$) is the $\ell^{3D}_{J/\psi}$ cut efficiency for prompt (non-prompt) J/ψ in Fig. 14. Table. 7 provides the calculated f_{prompt} for each p_T bin without $\ell^{3D}_{J/\psi}$ cut. Furthermore, prompt fraction can be easily estimated for J/ψ passing $\ell^{3D}_{J/\psi}$ cut, shown in Table. 8.

Table 7: The prompt J/ψ fraction f_{prompt} in pPb data without $\ell^{3D}_{J/\psi}$ cut.

p_T range (GeV/c)	f_{prompt}, $185 \leq N_{trk}^{offline} < 250$	f_{prompt}, $N_{trk}^{offline} < 35$
0.2–1.8	88.4%	83.8%
1.8–3.0	84.0%	82.8%
3.0–4.5	79.1%	79.8%
4.5–6.0	81.7%	79.0%
6.0–8.0	79.2%	74.9%
8.0–10.0	74.7%	70.6%

Table 8: The prompt J/ψ fraction f_{prompt} in pPb data with $\ell^{3D}_{J/\psi}$ cut.

p_T range (GeV/c)	f_{prompt}, $185 \leq N_{trk}^{offline} < 250$	f_{prompt}, $N_{trk}^{offline} < 35$
0.2–1.8	95.8%	93.9%
1.8–3.0	95.1%	94.7%
3.0–4.5	94.3%	94.5%
4.5–6.0	95.8%	95.0%
6.0–8.0	95.8%	94.7%
8.0–10.0	95.4%	94.4%

6 Analysis Technique

6.1 Dihadron correlation technique

The v_n harmonics for J/ψ particles are measured with a dihadron correlation technique follows exactly the same procedure as was established in Refs. [5, 6, 18], which was documented in detail in the previous analysis note, AN-12-352 (HIN-12-015). For each track multiplicity bin, trigger particles are defined as all J/ψ candidates with $1.4 < |y| < 2.4$ and in specified $p_{\mathrm{T}}^{\mathrm{trig}}$ range. The number of trigger particles in the event is denoted by N_{trig}, and there may be more than one trigger particle per event. Hadron pairs are formed by associating with every trigger particle the remaining charged particles with $|\eta| < 2.4$ and in a specified $p_{\mathrm{T}}^{\mathrm{assoc}}$ range. The per-trigger-particle associated yield distribution is then defined by:

$$\frac{1}{N_{\mathrm{trig}}}\frac{d^2N^{\mathrm{pair}}}{d\Delta\eta d\Delta\phi} = B(0,0) \times \frac{S(\Delta\eta,\Delta\phi)}{B(\Delta\eta,\Delta\phi)}, \tag{4}$$

where $\Delta\eta$ and $\Delta\phi$ are the differences in η and ϕ of the pair, respectively. The signal distribution, $S(\Delta\eta,\Delta\phi)$, is the measured per-trigger-particle distribution of same-event pairs, i.e.,

$$S(\Delta\eta,\Delta\phi) = \frac{1}{N_{\mathrm{trig}}}\frac{d^2N^{\mathrm{same}}}{d\Delta\eta d\Delta\phi}. \tag{5}$$

The mixed-event background distribution,

$$B(\Delta\eta,\Delta\phi) = \frac{1}{N_{\mathrm{trig}}}\frac{d^2N^{\mathrm{mix}}}{d\Delta\eta d\Delta\phi}, \tag{6}$$

is constructed by pairing the trigger particles in each event with the associated particles from 20 different random events, excluding the original event. The 20 random events are not required to contain a J/ψ candidate. The symbol N^{mix} denotes the number of pairs resulting from the event mixing. The normalization of both signal and background distributions by dividing by N_{trig} is done after summing up the pair density distributions for all the events and then dividing by the total number of trigger particles from all the events.

The background distribution is used to account for random combinatorial background and pair-acceptance effects. The normalization factor $B(0,0)$ is the value of $B(\Delta\eta,\Delta\phi)$ at $\Delta\eta = 0$ and $\Delta\phi = 0$ (with a bin width of 0.3 in $\Delta\eta$ and $\pi/16$ in $\Delta\phi$), representing the mixed-event associated yield for both particles of the pair going in approximately the same direction, thus having full pair acceptance. Therefore, the ratio $B(0,0)/B(\Delta\eta,\Delta\phi)$ is the pair-acceptance correction factor used to derive the corrected per-trigger-particle associated yield distribution. Equation (4) is calculated in 2 cm wide bins of the vertex position (z_{vtx}) along the beam direction and averaged over the range $|z_{\mathrm{vtx}}| < 15$ cm.

Each reconstructed track is weighted by the inverse of the efficiency factor, $\varepsilon_{\mathrm{trk}}(\eta, p_{\mathrm{T}})$, as a function of the track's pseudorapidity and transverse momentum. The efficiency weighting factor accounts for the detector acceptance $A(\eta, p_{\mathrm{T}})$, the reconstruction efficiency $E(\eta, p_{\mathrm{T}})$, and the fraction of misidentified tracks, $F(\eta, p_{\mathrm{T}})$,

$$\varepsilon_{\mathrm{trk}}(\eta, p_{\mathrm{T}}) = \frac{AE}{1-F}. \tag{7}$$

To further quantify the correlation structure, the 2-D distributions are reduced to one-dimensional (1-D) distributions in $\Delta\phi$ by averaging over the $\Delta\eta$ range [5, 6, 14, 18]. The azimuthal anisotropy harmonics are determined from a Fourier decomposition of long-range two-particle $\Delta\phi$ correlation functions,

$$\frac{1}{N_{\mathrm{trig}}}\frac{\mathrm{d}N^{\mathrm{pair}}}{\mathrm{d}\Delta\phi} = \frac{N_{\mathrm{assoc}}}{2\pi}\left\{1+\sum_{n} 2V_{n\Delta}\cos(n\Delta\phi)\right\}, \tag{8}$$

as described in Refs. [5, 6], where $V_{n\Delta}$ are the Fourier coefficients and N_{assoc} represents the total number of pairs per trigger particle for a given $(p_{\mathrm{T}}^{\mathrm{trig}}, p_{\mathrm{T}}^{\mathrm{assoc}})$ bin. The first three Fourier terms are included in the fits. Including additional terms have negligible effects on the fit results. A minimum η gap of 1 units is applied to remove short-range correlations from jet fragmentation. The elliptic anisotropy harmonics, $v_2\{2, |\Delta\eta| > 1\}$, of J/$\psi$ from two-particle correlation method can be extracted from the fitted Fourier coefficients as a function of p_{T},

$$v_n(p_{\mathrm{T}}^{\mathrm{trig}}) = \frac{V_{n\Delta}(p_{\mathrm{T}}^{\mathrm{trig}}, p_{\mathrm{T}}^{\mathrm{assoc}})}{\sqrt{V_{n\Delta}(p_{\mathrm{T}}^{\mathrm{assoc}}, p_{\mathrm{T}}^{\mathrm{assoc}})}}, n = 2, 3 \tag{9}$$

Here, the reference particle $p_{\mathrm{T}}^{\mathrm{ref}}$ range is chosen to be $0.3 < p_{\mathrm{T}} < 3.0\,\mathrm{GeV}/c$.

6.2 J/ψ v_n extraction

This analysis implement the similar technique to extract J/ψ v_n as described in AN-16-084 (HIN-16-007) and AN-17-223 (HIN-17-003), involving fit to the mass spectrum and v_n as function of invariant mass.

After v_n of all J/ψ candidates are measured as function of invariant mass, to extract v_n of J/ψ signal (v_n^{sig}), fits on mass spectrum and v_n as function of invariant mass is performed. The mass spectrum fit function is composed of two components: an exponential function to model the combinatorial background ($Bkg(m_{inv})$), a double Crystal Ball function for J/ψ signal ($Signal(m_{inv})$), as was done in AN-16-067 (HIN-16-004). The average v_n of all J/ψ candidates as a function of invariant mass, $v_n^{Sig+Bkg}(m_{inv})$, is fitted with Equation 10 and 11.

$$v_n^{Sig+Bkg}(m_{inv}) = \alpha(m_{inv})v_n^{sig} + (1 - \alpha(m_{inv}))v_n^{Bkg}(m_{inv}) \tag{10}$$

$$\alpha(m_{inv}) = Signal(m_{inv})/(Signal(m_{inv}) + Bkg(m_{inv})) \tag{11}$$

$v_n^{Bkg}(m_{inv})$ is the v_n of background J/ψ candidates and is modelled as a exponential function of invariant mass. $\alpha(m_{inv})$ is the J/ψ signal fraction as a function of invariant mass, which is from mass spectrum fit function. Systematic uncertainties from v_n fit is evaluated in Sec. 8.3.

Figure. 26 shows fit on mass spectrum and v_2 as function of invariant mass in different p_T bins for $185 \leq N_{trk}^{offline} < 250$.

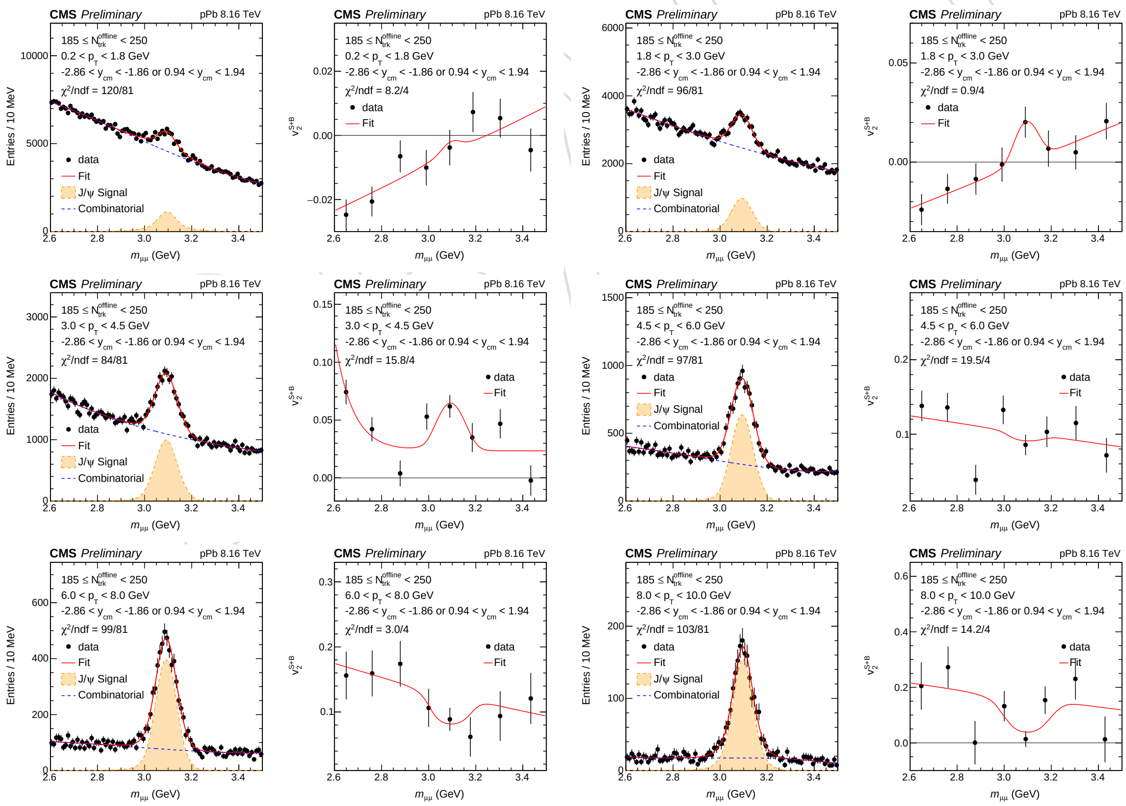

Figure 26: Fit on mass spectrum (left) and v_2 as function of invariant mass (right) in different p_T bins for $185 \leq N_{trk}^{offline} < 250$.

6.3 low multiplicity subtraction

The low multiplicity subtraction method developed in Refs. [17, 45] is employed in this analysis to estimate residual contribution from back-to-back jet correlations to v_2 results in pPb collisions. The v_2 results after subtracting correlations obtained from low-multiplicity pPb events are denoted as v_2^{sub}. The following paragraph provides brief description of the low multiplicity subtraction procedure, while more details can be found in AN-16-037 (HIN-16-010).

By assuming the jet-induced correlations are invariant with event multiplicity, the Fourier coefficients, $V_{n\Delta}$, extracted from Eq. 8 for $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$ are subtracted from the data in the higher multiplicity region:

$$V_{n\Delta}^{\mathrm{sub}} = V_{n\Delta} - V_{n\Delta}(N_{\mathrm{trk}}^{\mathrm{offline}} < 35) \times \frac{N_{\mathrm{assoc}}(N_{\mathrm{trk}}^{\mathrm{offline}} < 35)}{N_{\mathrm{assoc}}} \times \frac{Y_{\mathrm{jet}}}{Y_{\mathrm{jet}}(N_{\mathrm{trk}}^{\mathrm{offline}} < 35)}, \qquad (12)$$

where Y_{jet} represents the near-side jet yield which is defined as integral of near side jet peak in the short range ($|\Delta\eta| < 1$) 1-D $\Delta\phi$ correlation functions. The ratio, $Y_{\mathrm{jet}}/Y_{\mathrm{jet}}(N_{\mathrm{trk}}^{\mathrm{offline}} < 35)$, is introduced to account for the enhanced jet correlations due to the selection of higher multiplicity events.

In this analysis, the subtraction is performed with $V_{2\Delta}^{sig}$ which is obtained in the same way as for the v_2 results, as described in Sec. 6.2.

6.3.1 Jet yield estimation for J/Ψ-hadron correlation

The J/ψ candidates used in this analysis have $1.4 < |y| < 2.4$, which is at the edge or even beyond the $|\eta| < 2.4$ acceptance of charged particles. Therefore, charged particles which are in the same jets as J/ψ candidates can be outside the $|\eta| < 2.4$ acceptance, which leads to an in-complete near-side jet peak in the correlation functions, i.e. truncated.

The magnitude of this effect depends on the η value of J/ψ candidates. Figure. 27 shows the average η of J/ψ candidates in each p_{T} bin for events with $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$ and $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$, as well as their ratios. No strong multiplicity dependence has been observed for the entire p_{T} region measured, which indicates jets are more or less truncated by the same amount at low and high multiplicity. Thus, when taking the ratio of jet yield as in Eq. 13, the effect from truncation are largely cancelled.

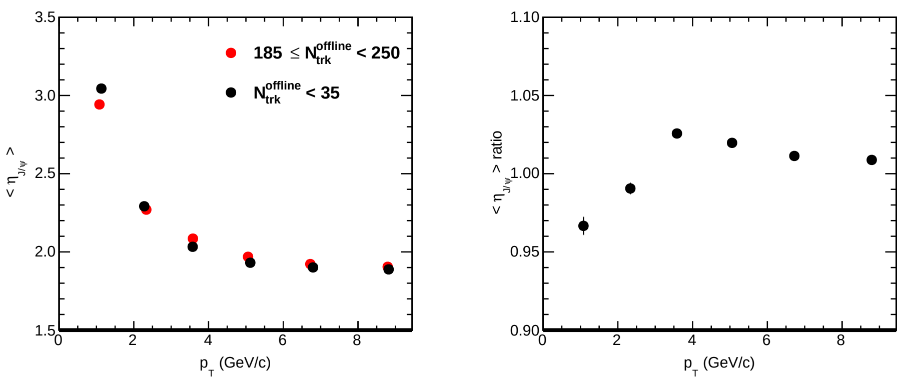

Figure 27: The average η of J/ψ candidates in each p_{T} bin for events with $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$ and $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$ (left), as well as their ratios (right).

To further ensure there is no large bias from truncated jet peak, jet yields are only extracted from correlation functions with J/ψ candidates have $p_{\mathrm{T}} > 4.5$ GeV/c, where no obvious jet peak truncation has been observed in the near-side 1D projection of the correlation functions, as shown in Fig. 28 and Fig. 29 for low and high multiplicity.

To extract the jet yield for signal J/ψ, a side-band method is implemented. As shown in Fig. 28 and Fig. 29, jet yields are evaluated in the background region ($2.55 < m_{\mu\mu} < 2.94$ combined with $3.24 < m_{\mu\mu} < 3.63$) and the peak region ($2.94 < m_{\mu\mu} < 3.24$) of the mass spectrum. Signal J/ψ meson jet yield is then calculated by

$$Y_{\mathrm{jet}}^{\mathrm{sig}} = (Y_{\mathrm{jet}}^{\mathrm{peak}} - (1-f)Y_{\mathrm{jet}}^{\mathrm{bkg}})/f, \tag{13}$$

where $Y_{\mathrm{jet}}^{\mathrm{peak}}$ and $Y_{\mathrm{jet}}^{\mathrm{bkg}}$ are jet yield from peak and background region, and f is the J/ψ signal fraction extracted from the mass spectrum in the peak region. The jet yield ratio $\frac{Y_{\mathrm{jet}}^{\mathrm{sig}}}{Y_{\mathrm{jet}}^{\mathrm{sig}}(N_{\mathrm{trk}}^{\mathrm{offline}}<35)}$ is then calculated as shown in Fig. 30, together with the same quantity for D^0 mesons (from HIN-17-003). Due to statistical limitations for J/ψ results and the observation that the same ratio for D^0 mesons has no strong p_{T} dependence, $\frac{Y_{\mathrm{jet}}^{\mathrm{sig}}}{Y_{\mathrm{jet}}^{\mathrm{sig}}(N_{\mathrm{trk}}^{\mathrm{offline}}<35)}$ is fitted by a straight line for $p_{\mathrm{T}} > 4.5 GeV$ for J/ψ. The constant from the fitting is found to be 1.61 ± 0.18 as shown in Fig. 30. The subtracted results with $\frac{Y_{\mathrm{jet}}^{\mathrm{sig}}}{Y_{\mathrm{jet}}^{\mathrm{sig}}(N_{\mathrm{trk}}^{\mathrm{offline}}<35)} = 1.61$ are reported as main results. To quoted systematic uncertainties for the subtraction, results are extracted with $\frac{Y_{\mathrm{jet}}^{\mathrm{sig}}}{Y_{\mathrm{jet}}^{\mathrm{sig}}(N_{\mathrm{trk}}^{\mathrm{offline}}<35)}$ varied by 2 sigma (i.e. 1.61 ± 0.36) in $p_{\mathrm{T}} > 4.5 GeV$ region and $\frac{Y_{\mathrm{jet}}^{\mathrm{sig}}}{Y_{\mathrm{jet}}^{\mathrm{sig}}(N_{\mathrm{trk}}^{\mathrm{offline}}<35)}$ varied by 3 sigma (i.e. 1.61 ± 0.54) in $p_{\mathrm{T}} < 4.5 GeV$ region. Figure. 31 shows v_2^{sub} results with different jet yield ratio, as well as the differences between them which are quote as systematic uncertainties. Table. 9 summarize the amount quoted.

Table 9: Summary of systematic uncertainties on v_2^{sub} from jet yield ratio for J/ψ in pPb collisions for $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$.

p_{T} range (GeV/c)	Systematic uncertainty
0.2–1.8	0.0023
1.8–3.0	0.0070
3.0–4.5	0.0071
4.5–6.0	0.0066
6.0–8.0	0.011
8.0–10.0	0.014

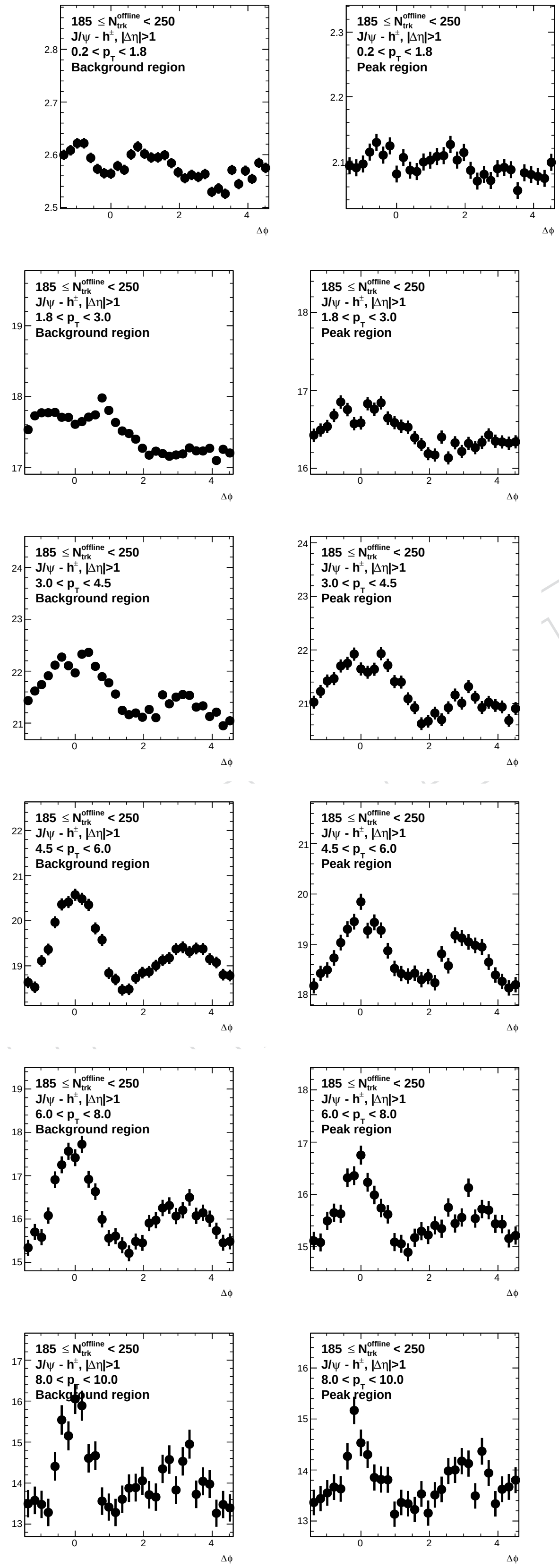

Figure 28: The near-side 1D projection of the J/ψ-hadron correlation functions for peak ($2.94 < m_{\mu\mu} < 3.24$) and background ($2.55 < m_{\mu\mu} < 2.94$ combined with $3.24 < m_{\mu\mu} < 3.63$) mass regions for $185 \leq N_{\rm trk}^{\rm offline} < 250$.

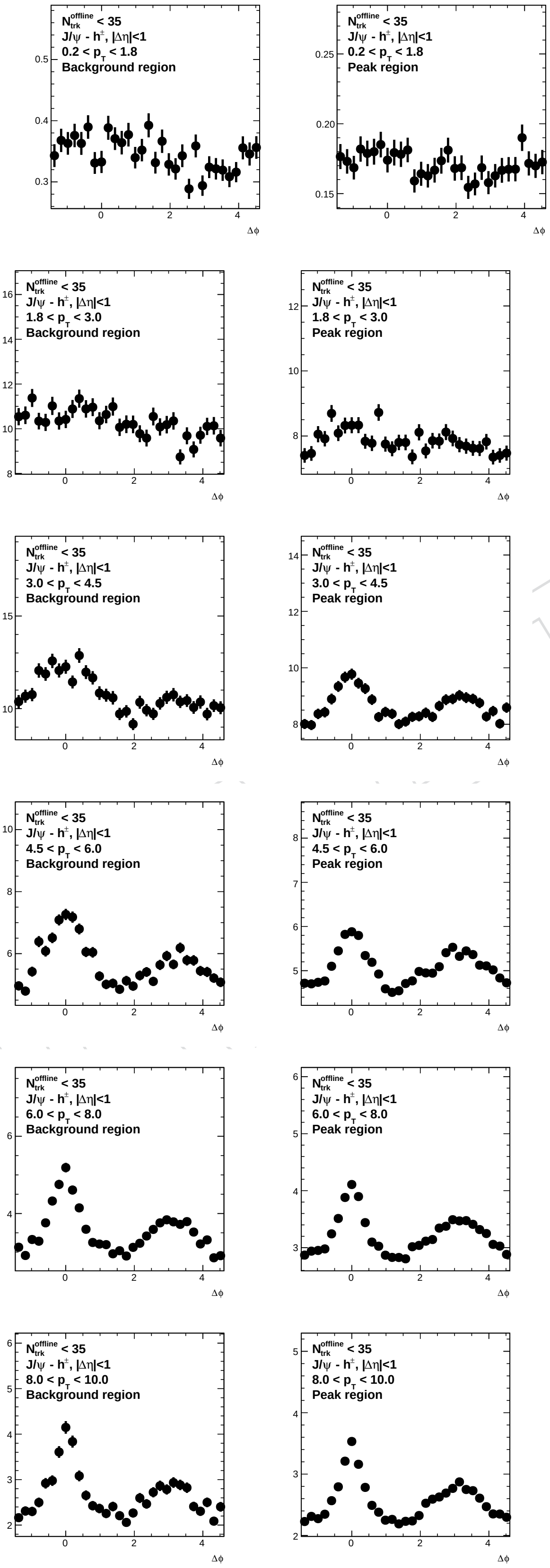

Figure 29: The near-side 1D projection of the J/ψ-hadron correlation functions for peak ($2.94 < m_{\mu\mu} < 3.24$) and background ($2.55 < m_{\mu\mu} < 2.94$ combined with $3.24 < m_{\mu\mu} < 3.63$) mass regions for $N_{\mathrm{trk}}^{\mathrm{offline}} < 35$.

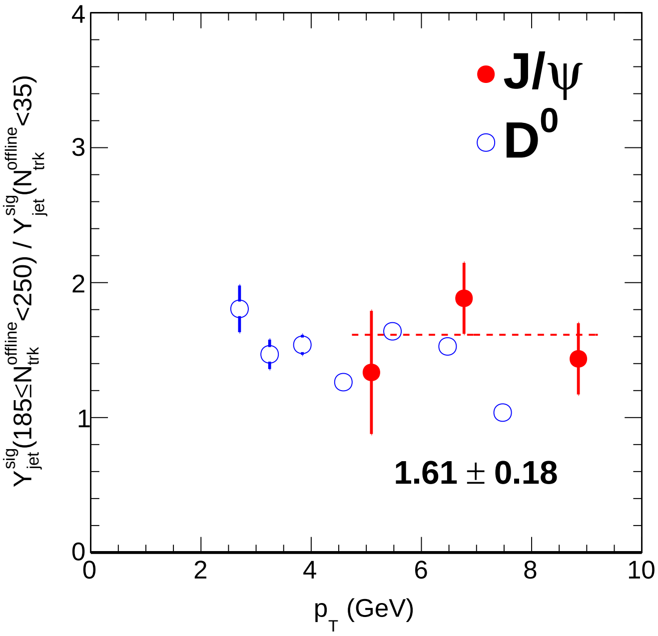

Figure 30: $\frac{Y^{sig}_{jet}}{Y^{sig}_{jet}(N^{offline}_{trk}<35)}$ for J/ψ (left) and D^0 (right) mesons.

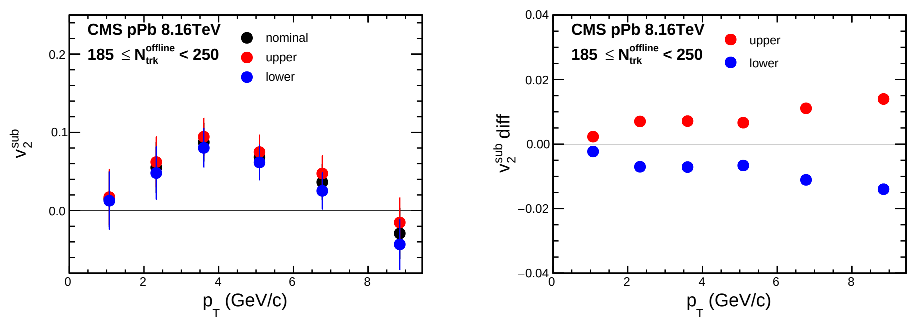

Figure 31: The v_2^{sub} results with different jet yield ratio, as well as the differences between results with jet yield ratio of 1 or 2 and results with jet yield ratio of 1.5.

7 Results

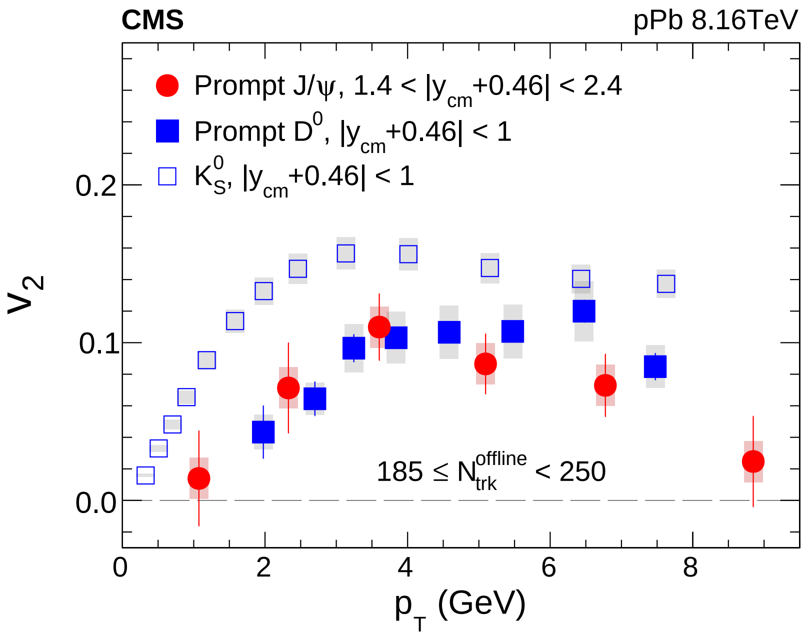

Figure 32: The v_2 results of the prompt J/ψ meson as a function of p_{T} in the multiplicity range $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$ of pPb collisions at $\sqrt{s_{NN}} = 8.16\,\mathrm{TeV}$. Data for $\mathrm{K}^0_{\mathrm{S}}$ and D^0 mesons from previous CMS measurements [46] are also shown for comparison. The error bars correspond to statistical uncertainties, while the shaded areas denote the systematic uncertainties.

Figure 32 shows the v_2 results of prompt J/ψ meson at forward rapidities ($-2.86 < y_{\mathrm{cm}} < -1.86$ and $0.94 < y_{\mathrm{cm}} < 1.94$) for high-multiplicity ($185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$) pPb collisions. Significant positive values of prompt J/ψ v_2 are observed over a wide p_{T} range from about 2 to 8 GeV. The prompt J/ψ v_2^{sub} results show a trend of rising and declining with p_{T}. This observed trend appears to be in common with other meson species, such as $\mathrm{K}^0_{\mathrm{S}}$ (light, strange flavor) and D^0 (open heavy flavor) mesons, reported in the previous CMS publication [46] (where several strange baryon species are also studied but not shown in this paper). At low and intermediate p_{T} range of $< 5\,\mathrm{GeV}$, the v_2 values for J/ψ and D^0 mesons are consistent with each other within uncertainties, while an indication of a smaller v_2 signal for J/ψ mesons than that of D^0 mesons is seen at higher p_{T} for $p_{\mathrm{T}} > 5\,\mathrm{GeV}$. Over the full p_{T} range, the v_2 signal for both J/ψ and D^0 mesons is smaller than that of $\mathrm{K}^0_{\mathrm{S}}$ mesons. This observation is consistent with earlier conclusion that charm quarks appear to develop a weaker collective dynamics than that of light quarks in small collision systems [46], unlike in AA collisions.

To better study the elliptic flow signal purely coming from long-range collective correlations, the v_2 results with residual jet correlations estimated and corrected for (v_2^{sub}) are shown in Fig. 33 (top) for prompt J/ψ mesons as well as for $\mathrm{K}^0_{\mathrm{S}}$ and D^0 mesons [46] as a function of p_{T} in pPb collisions with $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$. The effect of correction is most noticeable at very high p_{T} (although still not significant), while the overall p_{T} dependence of v_2 data remain unchanged. The correction for the v_2 of $\mathrm{K}^0_{\mathrm{S}}$ is larger (possibly because $\mathrm{K}^0_{\mathrm{S}}$ mesons are more coupled with the bulk multiplicity) and its v_2^{sub} value after the correction tend to converge with that of prompt J/ψ and D^0 at high p_{T}.

Motivated by the possible formation of a hydrodynamically expanding QGP medium in small systems, the elliptic flow signal for $\mathrm{K}^0_{\mathrm{S}}$, J/$\psi$ and D^0 mesons is compared as a function of trans-

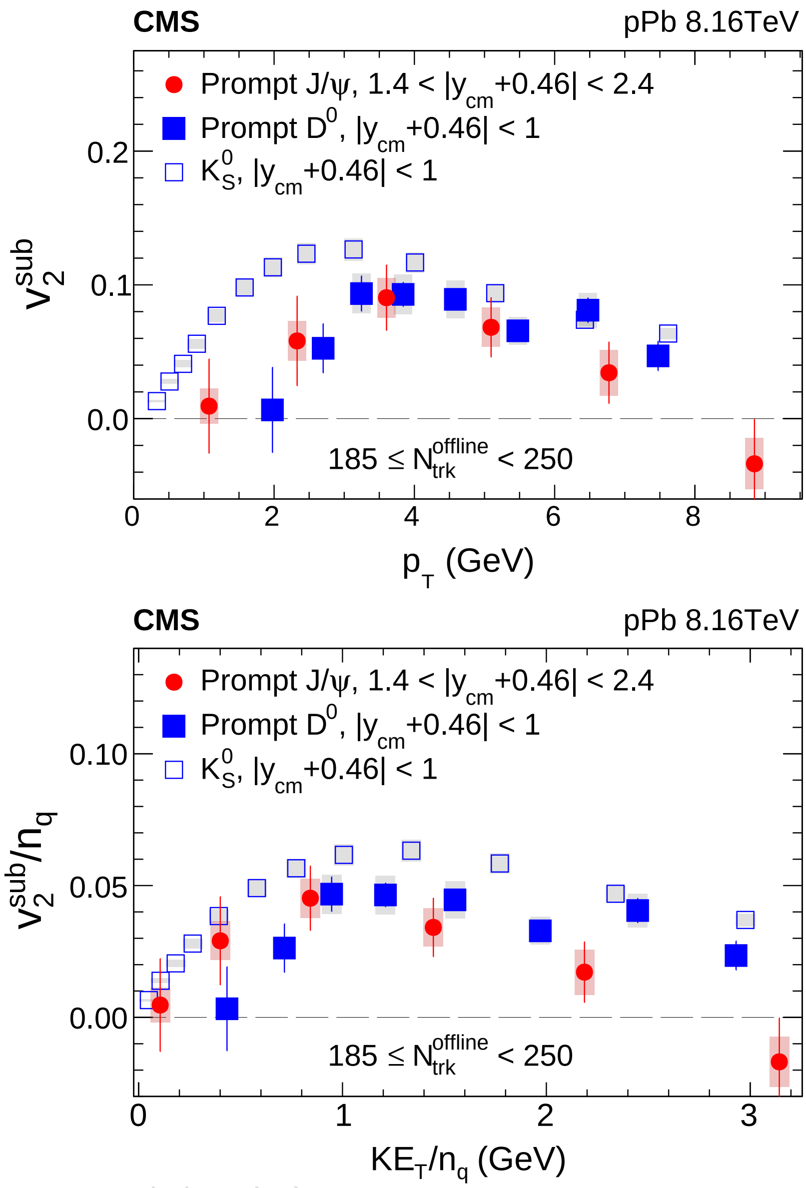

Figure 33: Top: the v_2^{sub} results for prompt J/ψ meson, as well as for $\mathrm{K^0_S}$ and prompt $\mathrm{D^0}$ mesons, as a function of p_{T} for pPb collisions at $\sqrt{s_{_{NN}}}$ = 8.16 TeV with $185 \leq N_{\mathrm{trk}}^{\mathrm{offline}} < 250$. Bottom: the n_{q}-scaled v_2^{sub} results. The $\mathrm{K^0_S}$ and $\mathrm{D^0}$ v_2 data are taken from Ref. [46]. The error bars correspond to statistical uncertainties, while the shaded areas denote the systematic uncertainties.

verse kinetic energy (KE_T) in Fig. 33 (bottom), to account for the mass difference among the three meson species [47, 48]. Here, the values of v_2^{sub} and KE_T are both divided by the number of constituent quarks, n_q, as shown in Fig. 33 (bottom), to represent collective flow signal at the partonic level, in the context of quark coalescence model [49–51]. A scaling of n_q-scaled v_2 values has observed between light baryon and meson species in pA [23, 46] and AA [47, 48, 52] systems (known as the NCQ scaling), indicating that collectivity is first developed among partons, which later recombine into final-state hadrons. However, testing the NCQ scaling is not the main goal in this Letter so only three meson species are compared with a common n_q value of 2. Over the full KE_T/n_q range, the elliptic flow signal per quark for prompt $\mathrm{D^0}$ and J/ψ mesons is consistently below that of $\mathrm{K^0_S}$ meson. For $KE_T/n_q > 1$ GeV, an indication of a smaller v_2 value of J/ψ than that of $\mathrm{D^0}$ is seen, which would again be consistent with a weaker collective behavior of heavy flavor quarks than light quarks or gluons, possibly due to the much smaller size of the collision system. However, no definitive conclusion can be drawn yet with current experimental uncertainties.